

GEORGE TSOUKALAS

Google DeepMind & University of Texas at Austin
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INDUSTRY & RESEARCH EXPERIENCE

Google DeepMind, Science and Strategic Initiatives Unit

Student Researcher, Computer Science Initiative

London, UK

May 2025 - April 2026

Trustworthy & Intelligent Systems Lab

Graduate Research Assistant

Austin, Texas

August 2023 - Present

- Developing neurosymbolic systems for formal mathematical reasoning.
- **Publication: (NeurIPS 2025 Spotlight)** Learning Interestingness in Automated Mathematical Theory Formation arXiv:2511.14778, November 2025.
- **Publication: (ICML AI4Math Best Paper Award, NeurIPS 2024)** PutnamBench: Evaluating Neural Theorem-Provers on the Putnam Mathematical Competition arXiv:2407.11214, July 2024.
- **Publication: (NeurIPS 2025)** CLEVER: A Curated Benchmark for Formally Verified Code Generation arXiv:2505.13938, May 2025.
- **Publication: (COLM 2024)** An In-Context Learning Agent for Formal Theorem Proving arXiv:2310.04353, March 2024.

Automated Reasoning Lab

Undergraduate Researcher

Rutgers University, New Jersey

August 2022 - May 2023

- Implemented neurosymbolic programming techniques with dPads and DifferentiableSygus using PyTorch.
- Researched syntax-guided synthesis of numerical invariants for program verification & synthesis.
- Utilized the Z3 Theorem Prover and Code2Inv framework for automated invariant checking.

Numerical Semigroups REU

Undergraduate Researcher

San Diego State University, California

May 2022 - August 2022

- Researched connections between numerical semigroups and a geometric object called the Kunz Cone.
- Uncovered several key properties regarding symmetries of the Kunz Cone and their action on unimaximal faces.
- Performed probabilistic analysis on Erdős-Rényi type random numerical semigroups towards Wilf's Conjecture.

Matrix Analysis REU

Undergraduate Researcher

College of William & Mary, Virginia

May 2021 - September 2021

- Researched generalized notions of graph adjacency matrices, including questions of eigenvalue assignments and disparity.
- Characterized k -NIM trees, extending a characterization of 1-NIM Trees by mentor Charles R. Johnson.
- Employed singularity analysis to enumerate k -NIM trees according to the newly found characterization.
- **Publication:** (Submitted) k -NIM Trees: Characterization and Enumeration, arXiv:2208.05450, July 2022.

EDUCATION

University of Texas at Austin

Ph.D. in Computer Science

Austin, Texas

August 2023 - Present

- **Research:** Third-year graduate student working on automated mathematical reasoning under Dr. Swarat Chaudhuri.

Rutgers University (Summa Cum Laude)

Bachelor of Science - Mathematics & Computer Science; Departmental GPA: 4.00/4.00

New Brunswick, New Jersey

September 2019 - May 2023

- **(2023) Kenneth and Rosalind Wolfson Award:** Awarded to an undergraduate senior majoring in mathematics based on academic merit.
- **(2022) Weill Scholarship:** Awarded to full-time students majoring in mathematics based on academic merit.
- **(2022) John Bogart Prize:** Awarded to a junior demonstrating excellence as a math major.
- **(2021) David Martin Weiss Memorial Award:** Awarded to a sophomore demonstrating excellence as a math major.
- **Honors College:** Recipient of the Rutgers Trustee & Henry Rutgers Scholarships.

PRESENTATIONS

- **Automated Theory Formation and Interestingness in Mathematics** - *Harvard CMSA Lecture Series*, February 2026.
- **Benchmarking Progress in Formal Mathematics** - *S-REPLs 2025, AWS London*, October 2025.
- **PutnamBench: A Multilingual Competition-Mathematics Benchmark for Formal Theorem-Proving** - *ICML AI for Math Workshop Best Paper*, July 2024.
- **Numerical Semigroups and the Kunz Cone** - *Joint Mathematics Meetings, Boston*, January 2023.
- **Geometric characteristics of symmetric numerical semigroups in the Kunz Cone** - *San Diego State*, August 2022.
- **Eigenvalues, Multiplicities, and Graphs** - *College of William & Mary*, July 2021.